

- 3 (a) The (mutual) electric force acting between two point charges is proportional to product of the charges and inversely proportional to the square of their separation.

- (b) Evidence of correct estimation of ΔV by area under graph between 3.3 cm to 5 cm

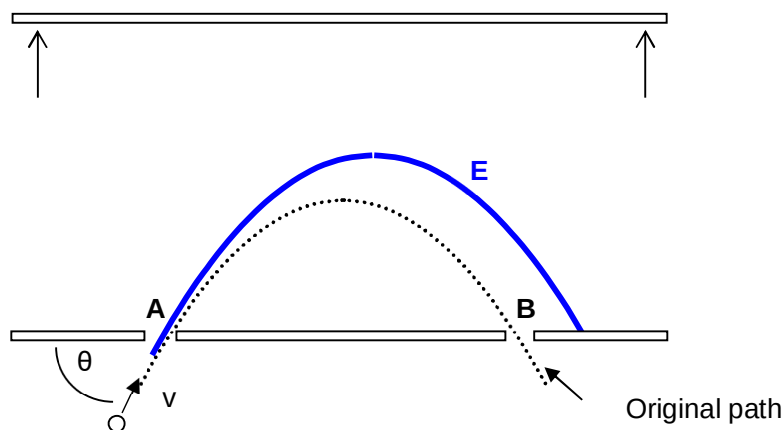
$$\begin{aligned}\text{Estimated } \Delta V &= 42 \text{ squares multiply by } (0.5 \times 10^3 \times 0.001) \text{ V per square} \\ &= 21 \text{ V}\end{aligned}$$

$$\begin{aligned}q \Delta V &= \frac{1}{2} m v^2 \\ (1.60 \times 10^{-19}) \times (21) &= \frac{1}{2} (1.67 \times 10^{-27}) v^2 \\ v &= 6.34 \times 10^4 \text{ m s}^{-1}\end{aligned}$$

(c) (i) $E = \frac{V - 100}{d}$

$$\begin{aligned}V &= Ed + 100 \\ &= (2500)(8.0 \times 10^{-2}) + 100 \\ &= 300 \text{ V}\end{aligned}$$

(ii)



Mark awarded only if parabolic shape and,
higher vertical displacement
further horizontal displacement

- (d) Volume, V , of each drop $\frac{4}{3} \pi r^3$, hence, $V \propto r^3$

After combining, the volume of new drop is $2V$, hence,

$$V \propto r^3 \quad \Rightarrow \quad \left(\frac{r'}{r}\right)^3 = \frac{2V}{V} \quad \Rightarrow \quad r' = (2^{1/3})r$$

Potential, $V = \frac{Q}{4\pi\epsilon_0 r}$, hence,

$$\begin{aligned}V \propto \frac{Q}{r} \quad \Rightarrow \quad \frac{V'}{V} &= \left(\frac{Q'}{Q}\right)\left(\frac{r}{r'}\right) = \left(\frac{2Q}{Q}\right)\left(\frac{r}{(2^{1/3})r}\right) \\ V' &= \frac{2}{(2^{1/3})}(500) = 793.7 \text{ V} \\ &= 794 \text{ V (3 sf)}\end{aligned}$$

4. (a) (i) Using Fleming Left hand rule, the magnetic force will always be perpendicular to